THE UNIVERSITY OF MELBOURNE

| Student number |
|---|

Semester 1 Final Exam, 2021

School of Mathematics and Statistics

# MAST10012 Introduction to Mathematics

Reading time: 30 minutes — Writing time: 3 hours — Upload time: 30 minutes

This exam consists of 20 pages (including this page)

## Permitted Materials

- This exam and/or an offline electronic PDF reader, one or more copies of the masked exam template made available earlier and blank loose-leaf paper.
- No books, notes or other printed or handwritten material are permitted.
- No calculators are permitted.

## Instructions to Students

- If you have a printer, print the exam one-sided. If using an electronic PDF reader to read the exam, it must be disconnected from the internet. Its screen must be visible in Zoom. No mathematical or other software on the device may be used. No file other than the exam paper may be viewed.
- Ask the supervisor if you want to use the device running Zoom.

*Writing*

- There are 10 questions with marks as shown. The total number of marks available is 110.
- Write your answers in the boxes provided on the exam that you have printed or the masked exam template that has been previously made available. If you need more space, you can use blank paper. Note this in the answer box, so the marker knows. The extra pages can be added to the end of the exam to scan.
- If you have been unable to print the exam and do not have the masked template write your answers on A4 paper. The first page should contain only your student number, the subject code and the subject name. Write on one side of each sheet only. Start each question on a new page and include the question number at the top of each page.

*Scanning*

- Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Scan directly from above. Crop pages to A4. Make sure that you upload the correct PDF file and that your PDF file is readable.

*Submitting*

- **You must submit while in the Zoom room.** No submissions will be accepted after you have left the Zoom room.
- Go to the Gradescope window. Choose the Canvas assignment for this exam. Submit your file. Wait for Gradescope email confirming your submission. Tell your supervisor when you have received it.

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**Question 1 (27 marks)**

**Short-answer questions:** Full marks will be given for a correct answer placed in the box. Part marks will be given for incorrect answers if working is shown.

(a) In how many different ways can the letters of the word 'DETAIL' be arranged such that the vowels must occupy only the odd positions? (assume position numbers to be 1-6)

(b) Evaluate $\cos\left(\frac{3\pi}{2}\right) + \sin\left(-\frac{\pi}{6}\right)$

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(c) Let $f(x) = (x + \sin(x)) \log(x)$. Find $f'(x)$.

(d) Let $g(x) = \sqrt{e^x + 6}$. Find $g'(x)$.

(e) Let $h(x) = \dfrac{x^3 + 2x^2 + \frac{6}{x}}{e^x}$. Find $h'(x)$.

(f) Find the equation of the tangent line of $f(x) = e^{\cos(x)}$ about $x = \frac{\pi}{2}$

(g) Evaluate the following:

$$\int \frac{7x^4 + 3x}{x^5} \, dx$$

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(h) Evaluate the following:

$$\int_0^2 \left(\sin(\pi x) + \sqrt[5]{x}\right) \; dx$$

(i) Solve the following initial value problem

$$F(x) = \int \frac{7}{x} \; dx \quad \text{if } F(e) = 1$$

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**Full-solution problems:** Complete solutions are required for the remainder of this examination. Be sure to justify your answers and to show all your work. In addition to the accuracy of your answer and methods, you will be assessed on the justifcation of your solutions.

**Question 2 (7 marks)**

Find all solutions of the trigonometric equation $2\sin\left(\frac{\pi x}{2}\right) + 1 = 0$.

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**Question 3 (7 marks)**

(a) Evaluate the following limit if it exists. Otherwise, write DNE and justify why the limit does not exist.

$$\lim_{x \to 1} \frac{x^3 + 2x^2 - x - 2}{x(x-1)}$$

(b) Use the limit definition of the derivative to find $f'(x)$ given $f(x) = \frac{1}{2}x^3$. No marks will be awarded for using any other method.

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**Question 4 (15 marks)**

Define the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = -3x^2 + 12$.

(a) What is the turning point of $f$?

(b) What is the domain, codomain and range of $f$?

(c) Is $f$ injective? Is $f$ surjective?

(d) Find the largest value $q$ such that the function $f : (-\infty, q] \to \mathbb{R}$ given by $f(x) = -3x^2 + 12$ is injective.

(e) Find the interval $C$ such that $f : (-\infty, q] \to C$ given by $f(x) = -3x^2 + 12$ is surjective.

(f) Based on your choice of $q$ and $C$, explain why the function $f : (-\infty, q] \to C$ given by $f(x) = -3x^2 + 12$ is invertible.

(g) Find the inverse of $f$ and state its domain and codomain.

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**Question 5 (11 marks)**

Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = (x-1)^2(x+2)$.

(a) Find all $x$ and $y$ intercepts of the function.

(b) Find the intervals where $f(x)$ is positive and intervals where $f(x)$ is negative.

(c) Evaluate and simplify $f'(x)$.

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(d) Find the points in the domain where $f'(x) = 0$.

(e) Find the intervals where $f(x)$ is increasing and intervals where $f(x)$ is decreasing.

(f) Find and classify the location of any local extrema of the function.

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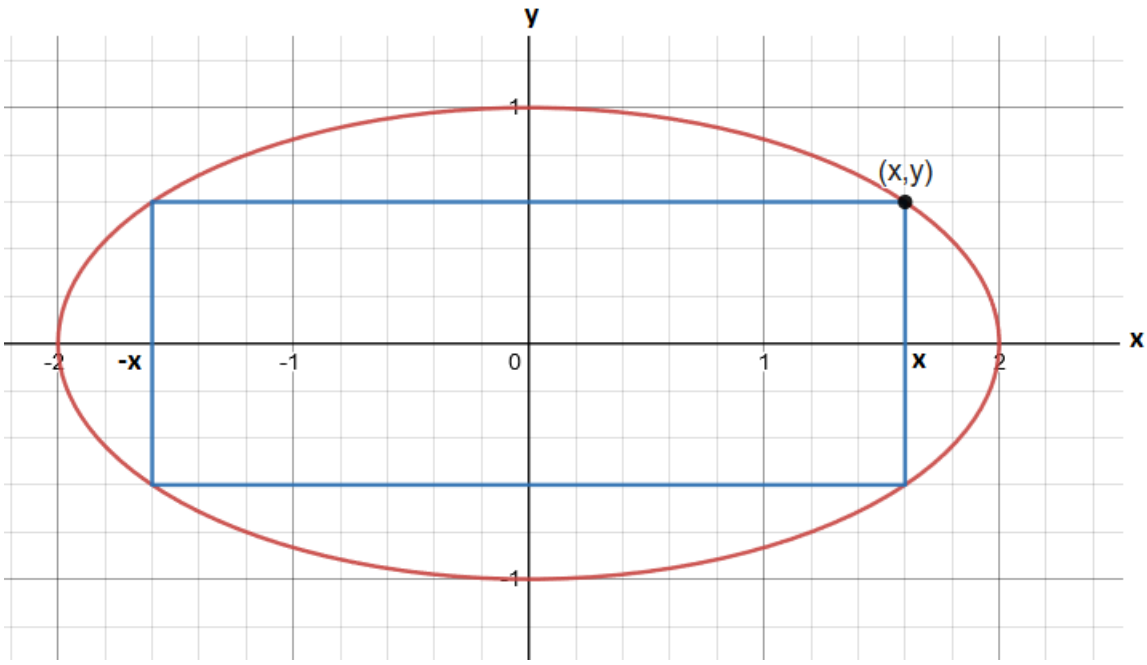
**Question 6 (12 marks)**

A rectangle is to be inscribed in the ellipse

$$\frac{x^2}{4} + y^2 = 1$$

in the following manner:




(a) What is the length of the base of the rectangle in terms of $x$?

(b) Given that the vertices of the rectangle touch the ellipse, what is the height of the rectangle in terms of $x$?

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(c) Write down an expression for the area of the rectangle in terms of $x$. What is the domain of $A(x)$?

(d) Determine the dimensions (length, height and area) of the rectangle with the maximum area that will fit in this ellipse.

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**Question 7 (10 marks)**

Consider two functions given by $y = x^2 - 5$ and $y = 3x - 5$.

(a) Find the coordinates where the two functions intersect

(b) Compute the physical area enclosed by the two functions.

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**Question 8 (5 marks)**

*For this question, a numerical answer is not required. You may leave your final answer in simplified fractional form.*

Roll a die 10 times and observe the number you get each time.

(a) What is the probability that you roll an odd number exactly 3 times?

(b) What is the probability that you roll an odd number at least 8 times?

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**Question 9 (9 marks)**

The height of a plant called *cryptomeria japonica* is normally distributed with a mean of 100 cm and a standard deviation of 5 cm.

(a) What is the probability that a cryptomeria japonica chosen at random would be:

(i) Shorter than 110cm?

(ii) Between 95cm to 110cm?

(b) What is the minimum height for a cryptomeria japonica to be among the tallest 10% of the species?

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**Question 10 (7 marks)**

(a) Plot the function $f : [-1, 1) \to \mathbb{R}$, $f(x) = \frac{x+1}{2}$.

(b) Show that $f$ from part (a) is a probabilty distribution.

**End of Exam — Total Available Marks = 110**

**Turn the page for appended material**

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# Formula Sheet

- **Combinatorics:**

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot 3 \cdot 2 \cdot 1$$
$$\binom{n}{k} = \frac{n!}{(n-k)!k!}$$

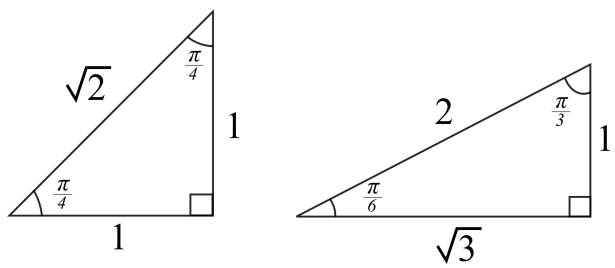
- **Functions:**
  - Lines

$$y = mx + c \qquad\qquad y - b = m(x-a)$$

  - Quadratics

$$y = ax^2 + bx + c = a(x-r)(x-s) = a(x-h)^2 + k$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- **Special Triangles:**




- **Trigonometric identities:**

$$\sin(x + 2\pi) = \sin(x) \qquad \cos(x + 2\pi) = \cos(x)$$
$$\sin(\pi - x) = \sin(x) \qquad \sin(-x) = -\sin(x)$$
$$\cos(\pi - x) = -\cos(x) \qquad \cos(-x) = \cos(x)$$

$$\sin(x+y) = \sin(x)\cos(y) + \cos(x)\sin(y) \qquad \cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$$
$$\sin(2x) = 2\sin(x)\cos(x) \qquad \cos(2x) = \cos^2(x) - \sin^2(x)$$
$$\sin^2(x) = \frac{1 - \cos(2x)}{2} \qquad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

- **Exponential and logarithms:**

$$a^0 = 1 \qquad a^1 = a$$
$$a^{-1} = \frac{1}{a} \qquad a^{\frac{1}{2}} = \sqrt{a}$$
$$a^x \cdot a^y = a^{x+y} \qquad \frac{a^x}{a^y} = a^{x-y}$$
$$(a^x)^y = a^{x \cdot y} \qquad \sqrt[y]{a^x} = a^{\frac{x}{y}}$$
$$a^x \cdot b^x = (ab)^x \qquad \frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$$

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$$\log_e(x) = \ln(x) \qquad \log_a(1) = 0$$
$$\log_a(a) = 1 \qquad \log_a(p^x) = x\log_a(p)$$
$$\log_a(pq) = \log_a(p) + \log_a(q) \qquad \log_a\left(\frac{p}{q}\right) = \log_a(p) - \log_a(q)$$
$$\log_a(p) = \frac{\log_c(p)}{\log_c(a)}$$

- **Limits:** If we have

$$\lim_{x\to a} f(x) = K \qquad \lim_{x\to a} g(x) = L$$

Then

$$\lim_{x\to a}[f(x)+g(x)] = K+L \qquad \lim_{x\to a}[f(x)-g(x)] = K-L$$
$$\lim_{x\to a}[f(x)\times g(x)] = K\times L \qquad \lim_{x\to a}\left[\frac{f(x)}{g(x)}\right] = \frac{K}{L}$$
$$\lim_{x\to a}[f(x)]^n = \left[\lim_{x\to a} f(x)\right]^n \qquad \lim_{x\to a}[f(x)]^{\frac{1}{n}} = \left[\lim_{x\to a} f(x)\right]^{\frac{1}{n}}$$
$$\lim_{x\to a} cf(x) = cK$$
$$\lim_{x\to a} c = c \qquad \lim_{x\to a} x = a$$
$$\lim_{x\to a} x^n = a^n \qquad \lim_{x\to a} x^{\frac{1}{n}} = a^{\frac{1}{n}}$$

- **Derivatives:**

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{a\to x}\frac{f(a)-f(x)}{a-x}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(c\Big) = 0 \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(x^k\Big) = kx^{k-1}$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\sqrt{x}\Big) = \frac{1}{2\sqrt{x}} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(e^x\Big) = e^x$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\log(x)\Big) = \frac{1}{x} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(\sin(x)\Big) = \cos(x)$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\cos(x)\Big) = -\sin(x) \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(\tan(x)\Big) = \frac{1}{\cos^2(x)}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(u(x)\cdot v(x)\Big) = \frac{\mathrm{d}u}{\mathrm{d}x}\cdot v(x) + u(x)\cdot\frac{\mathrm{d}v}{\mathrm{d}x} = u'(x)\cdot v(x) + u(x)\cdot v'(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u(x)}{v(x)}\right) = \frac{\frac{\mathrm{d}u}{\mathrm{d}x}\cdot v(x) - u(x)\cdot\frac{\mathrm{d}v}{\mathrm{d}x}}{[v(x)^2]} = \frac{u'(x)\cdot v(x) - u(x)\cdot v'(x)}{[v(x)^2]}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(f(u(x))\Big) = \frac{\mathrm{d}f}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}x} = f'(u)\cdot u'(x)$$

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**The standard normal distribution table for $P(X < x)$**

| $x$ | **0.00** | **0.01** | **0.02** | **0.03** | **0.04** | **0.05** | **0.06** | **0.07** | **0.08** | **0.09** |
|---|---|---|---|---|---|---|---|---|---|---|
| **0.0** | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| **0.1** | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| **0.2** | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| **0.3** | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| **0.4** | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| **0.5** | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| **0.6** | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| **0.7** | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| **0.8** | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| **0.9** | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| **1.0** | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| **1.1** | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| **1.2** | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| **1.3** | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| **1.4** | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| **1.5** | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| **1.6** | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |
| **1.7** | 0.9554 | 0.9564 | 0.9573 | 0.9582 | 0.9591 | 0.9599 | 0.9608 | 0.9616 | 0.9625 | 0.9633 |
| **1.8** | 0.9641 | 0.9649 | 0.9656 | 0.9664 | 0.9671 | 0.9678 | 0.9686 | 0.9693 | 0.9699 | 0.9706 |
| **1.9** | 0.9713 | 0.9719 | 0.9726 | 0.9732 | 0.9738 | 0.9744 | 0.9750 | 0.9756 | 0.9761 | 0.9767 |
| **2.0** | 0.9772 | 0.9778 | 0.9783 | 0.9788 | 0.9793 | 0.9798 | 0.9803 | 0.9808 | 0.9812 | 0.9817 |
| **2.1** | 0.9821 | 0.9826 | 0.9830 | 0.9834 | 0.9838 | 0.9842 | 0.9846 | 0.9850 | 0.9854 | 0.9857 |
| **2.2** | 0.9861 | 0.9864 | 0.9868 | 0.9871 | 0.9875 | 0.9878 | 0.9881 | 0.9884 | 0.9887 | 0.9890 |
| **2.3** | 0.9893 | 0.9896 | 0.9898 | 0.9901 | 0.9904 | 0.9906 | 0.9909 | 0.9911 | 0.9913 | 0.9916 |
| **2.4** | 0.9918 | 0.9920 | 0.9922 | 0.9925 | 0.9927 | 0.9929 | 0.9931 | 0.9932 | 0.9934 | 0.9936 |
| **2.5** | 0.9938 | 0.9940 | 0.9941 | 0.9943 | 0.9945 | 0.9946 | 0.9948 | 0.9949 | 0.9951 | 0.9952 |
| **2.6** | 0.9953 | 0.9955 | 0.9956 | 0.9957 | 0.9959 | 0.9960 | 0.9961 | 0.9962 | 0.9963 | 0.9964 |
| **2.7** | 0.9965 | 0.9966 | 0.9967 | 0.9968 | 0.9969 | 0.9970 | 0.9971 | 0.9972 | 0.9973 | 0.9974 |
| **2.8** | 0.9974 | 0.9975 | 0.9976 | 0.9977 | 0.9977 | 0.9978 | 0.9979 | 0.9979 | 0.9980 | 0.9981 |
| **2.9** | 0.9981 | 0.9982 | 0.9982 | 0.9983 | 0.9984 | 0.9984 | 0.9985 | 0.9985 | 0.9986 | 0.9986 |
| **3.0** | 0.9987 | 0.9987 | 0.9987 | 0.9988 | 0.9988 | 0.9989 | 0.9989 | 0.9989 | 0.9990 | 0.9990 |
| **3.1** | 0.9990 | 0.9991 | 0.9991 | 0.9991 | 0.9992 | 0.9992 | 0.9992 | 0.9992 | 0.9993 | 0.9993 |
| **3.2** | 0.9993 | 0.9993 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9995 | 0.9995 | 0.9995 |
| **3.3** | 0.9995 | 0.9995 | 0.9995 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9997 |
| **3.4** | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9998 |
| **3.5** | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 |
| **3.6** | 0.9998 | 0.9998 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 |
| **3.7** | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 |
| **3.8** | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 |
| **3.9** | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |